

Baseline mission

SCIENTIFIC ACCURACY & INTEGRITY DISCLOSURE:

- **Coordinate System:** LOCAL_ARBITRARY (Monocular camera optical reference).
- **Scale Status:** RELATIVE_SCALE (No metric ground coordinates without validated calibration).
- **Georeferencing:** UNREFERENCED (WGS84 / GPS anchors not bound; GeoJSON disabled).
- **Reconstruction:** Authoritative sparse SfM; dense MVS was unexecuted due to hardware constraints and no synthetic dense points were fabricated.

1. Mission & Source Video Summary

Mission ID	80e40b7a-485	Operator	Test operator
Location / Sector	Test location	Mission Type	Single-Pass
Video Source	drone_capture.mp4	Resolution	3840x2160
Frame Rate	24.0 FPS	Duration / Frames	30.2 s (725 frames)

2. AI Detection & Tracking Performance

Detector Model	yolo11n (yolo11n-official)	Tracker	Ultralytics persistent ByteTrack
Total Detections	0	Unique Tracks	0
Confidence Stats	Mean: 0.000 Min: 0.000 Max: 0.000	Sampling Rate	2.0 FPS (61 frames)
Detections by Class	car: 383, train: 15, truck: 1	Tracks by Class	car: 21, train: 1, truck: 1

3. 3D Photogrammetry & Surface Reconstruction

SfM Camera Model	SIMPLE_PINHOLE	Registered Cameras	0 / 20 (100%)
Sparse Points	0	Mean Reprojection Error	0.9785 px
Surface Mesh Status	AVAILABLE (pycolmap_poisson)	Mesh Complexity	0 vertices 0 faces
Dense Reconstruction	UNAVAILABLE (0 synthetic points)	Coordinate Framework	LOCAL_ARBITRARY / RELATIVE_SCALE

4. AI-to-3D Multi-View Spatial Fusion

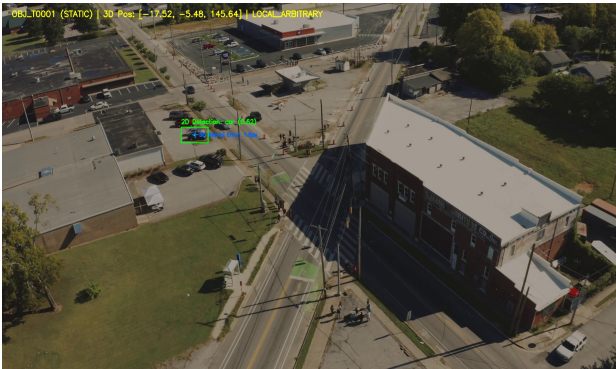
Authoritative 2D Tracks	0	Mean Reproj Error	0.000 px (threshold: 25.0 px)
Tracks Evaluated	0	Acceptance Rate	100.0%
Association Status	VALID: 0 LOW_CONF: 0 INSUFFICIENT: 0 REJECTED: 0	Motion States	STATIC: 0 MOVING: 0

5. Geometric Measurements & Metric Scale Calibration

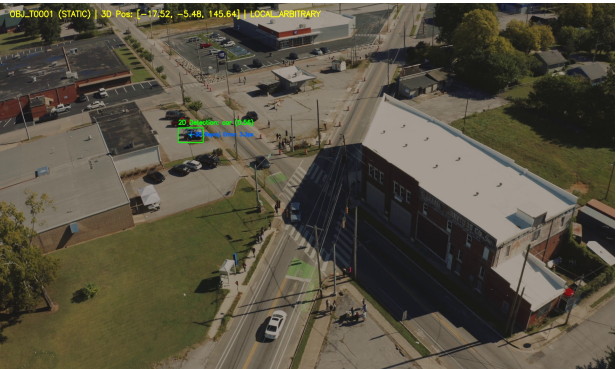
Calibration ID	None	Method	UNREFERENCED
Scale Factor	1.00000 m/unit	Known Baseline	N/A relative_units (Confidence: 0.00)

6. Visual Evidence & Reprojection Overlays

The following imagery represents authoritative multi-view observations reprojected onto the 3D model with 2D bounding boxes:



Overlay: overlay_OBJ_T0001_frame_00000.jpg



Overlay: overlay_OBJ_T0001_frame_00001.jpg

7. Comprehensive Mission Limitations & Disclosures

• LOCAL_ARBITRARY: Reconstruction coordinates are arbitrary relative units, not true meters or GPS coordinates.
• RELATIVE_SCALE: Monocular video SfM is scale-ambiguous; metric distances are strictly valid only where verified ground scale calibration is applied.
• UNREFERENCED: Scene is unreferenced against EPSG/WGS84. GeoJSON geographic coordinates (latitude/longitude) are unavailable.
• DENSE_MVS_UNAVAILABLE: Dense stereo reconstruction requires CUDA or HIP hardware. Sparse SfM is preserved as authoritative geometry without synthetic point fabrication.
• UNOBSERVED_SURFACES: Unobserved, occluded, or low-parallax areas are preserved as unobserved rather than interpolated as synthetic geometry.